LOCALIZED PHOTONIC NANOJET ENABLED QUANTITATIVE OPTICAL ENHANCEMENT

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ABSTRACT

Method of developing quantitative optical enhancement through photonic nanojet (PNJ) is previously undescribed. In this work, we localize PNJ in a semi-open microwell by utilizing dielectric microsphere based optical microstructures. We demonstrate that individual fluorescent microsphere can be introduced to the designated position in the PNJ and generates reproducible enhanced fluorescence signals, achieving quantitative optical enhancement.

KEYWORDS: Photonic Nanojet, Field Enhancement, Quantitative Enhancement, Enhancement Factor

INTRODUCTION

The convergence and confinement of light can efficiently deliver optical energy to micro- and nano-scale objects and enable strong light–matter interactions, which has a profound effect on the efficacy of many optical processes [1,2]. In recent years, dielectric microbeads have been shown their capability of converging light to the subwavelength scale through generation of PNJ without requiring expensive nanofabrication facilities or complex near-field configurations [3]. However, a key challenge is to introduce the object of interest into the extremely small volume of PNJ and induce reliable optical enhancement that benefits the signal quantitation. It requires precise control of the position of objects in the PNJ. Here, we present a dielectric microbead based optical microstructure which localizes PNJ in a semi-open microwell and allows passive trapping of individual fluorescent microsphere into the PNJ with high degree of spatial accuracy. Reproducible enhanced fluorescence signals can be induced, allowing quantitative measurement of signals amplified by enhanced light–matter interactions through PNJ for the first time.

EXPERIMENTAL

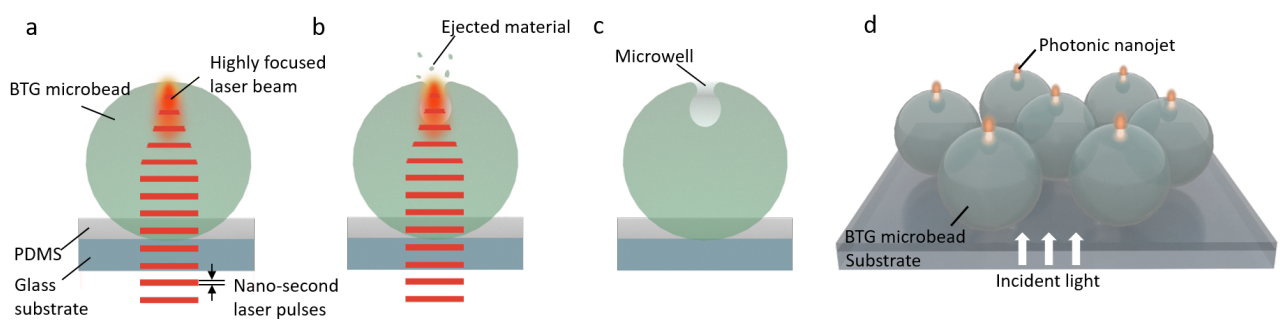

Figure 1. a-c. Schematic illustration of the fabrication process of the semi-open microwell in the BTG microbead. d. Upon illumination, the PNJ is generated in the semi-open microwell.

The fabrication process is schematically illustrated in Figure 1. Barium titanate glass (BTG) dielectric microbeads (refractive index 1.9) with size of ~80 µm are coated on a glass substrate with a layer of polydimethylsiloxane (PDMS). Upon exposing to a nanosecond laser from the bottom of the glass substrate, a focused high-intensity beam is generated near the top surface inside the BTG microbeads. Due to the heating effect generated by the highly focused laser beam, the illuminated BTG material is melted and ejected from the BTG microbeads, resulting a single semi-open microwell on each BTG microbead.

RESULTS AND DISCUSSION

Scanning electron microscope (SEM) image shows that a larger outer contour is generated on the top of the BTG microbead and the width of the contour gradually decreases and eventually narrows into a semi-open microwell with a diameter of about 6μm (Figure 2a and b). The interior of the semi-open microwell is observed as an ellipsoidal cavity. Fluorescent microspheres of 5 μm as target in dispersion can be passively trapped inside the semi-open microwell driven by the capillary force as the solvent gradually evaporates. Microwells can be filled at a yield approaching 70% after 5 loading steps. Besides, once the microsphere is trapped, it is automatically exposed in the enhanced electromagnetic field (PNJ) without further manipulation (Figure 2d). This optical microstructure greatly improves the efficiency of introducing the targets into the PNJs as well as suppresses spatial variations of the targets in the semi-open microwells, allowing reliable measurement of amplified fluorescent signals (Figure 2e). Fluorescent microspheres inside the semi-open microwell exhibit higher emission intensity than those on the substrate, Figure 2f and g. About 6.4-fold of enhancement is obtained.

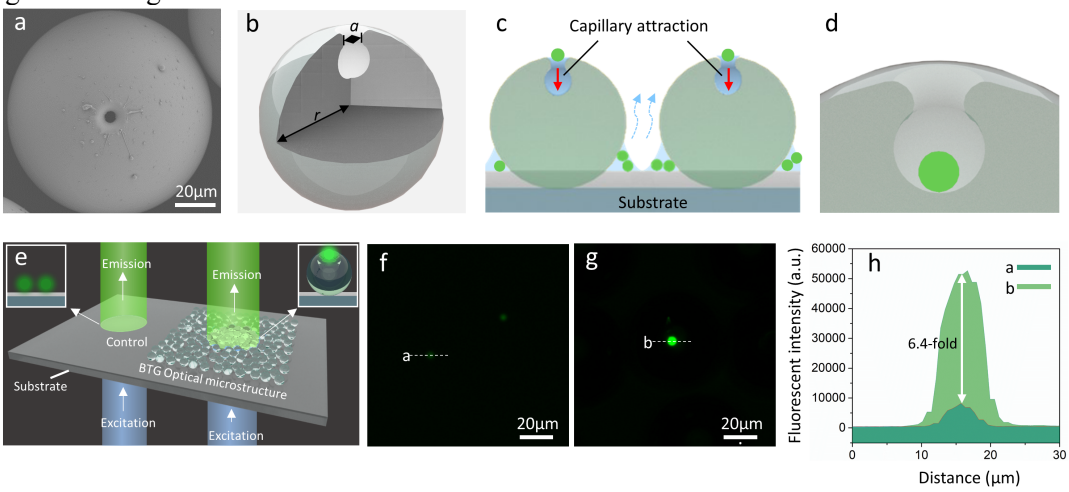

Figure 2. a. SEM image of the optical microstructure. b. Schematic illustration of the dimensions of the BTG microbead with the semi-open microwell. c. Schematics depicting the fluorescent microsphere trapped inside the semi-open microwell. d. Schematic illustration of the fluorescent microsphere of 5μm loaded in the semi-open microwell. e. Schematic illustration of the signal recording for the fluorescent microspheres on the substrate (control) and in the semi-open microwells. f and g. Corresponding recorded fluorescent signals of the microspheres on the substrate and in the microwell, respectively. h. Intensity of the corresponding fluorescent microspheres showing the enhancement (6.4-fold) of the fluorescent signals.

CONCLUSION

The optical microstructure demonstrated here will assist in realizing on-chip signal amplification and quantitative detection systems with high sensitivity and low detection limit, as well as providing a platform for investigating a wide range of light–matter interaction processes.

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